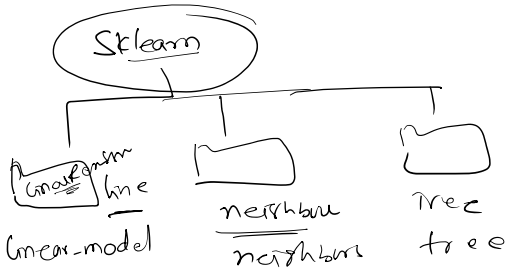
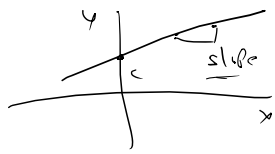
AIML-27-LinearRegression-KNN

16 May 2026 18:25

LinearRegression → sklearn



$y = mx + c$



from sklearn.linear_model import LinearRegression

step1: import all required libraries

step2: read the data into dataframe

step3: preprocess data → nullvalues
noise data
data type mismatch
} resolve

step4: Scaling → standard scaling x = ?
min max " y = ?
Robust "
Normalization

step5: we need to define
split the data x_train and y_train

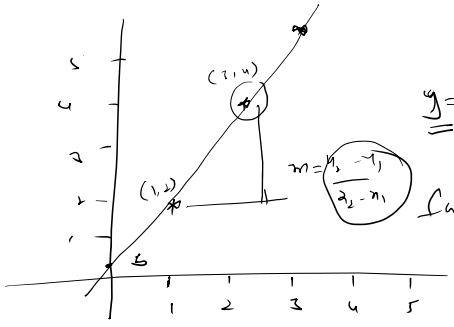
step6: create object of algorithm

step7: fit x_train, y_train

step8: find the accuracy

step9: predict for unknown data

x	y
1	2
2	4
3	5



$y = b + mx$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$f(x) = y = b + mx$

$y = b + mx$

for $x = 1$

$2 = 1b + m.1$

1	2
2	4
3	5

$x = 2$ | $y = 1.b + m.2$

$x = 3$ | $y = 1.b + m.3$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

$b + m = 2$

$b + 2m = 4$

$b + 3m = 5$

$$X\,\theta = y$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \cdot \theta = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}}_{X} \underbrace{\begin{bmatrix} b \\ m \end{bmatrix}}_{\theta} = \underbrace{\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}}_{y}$$

b, m

Ordinary least square method

$$\theta = (X^T . X)^{-1} (X^T y) \Rightarrow (X^T . X)\,\theta = X^T y$$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \qquad y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

$\downarrow$ Transpose

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$X^T . X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1+1 & 1+2+3 \\ 1+2+3 & 1+4+9 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$X^T . y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2+4+5 \\ 2+8+15 \end{bmatrix} = \begin{bmatrix} 11 \\ 25 \end{bmatrix}$$

$$(X^T . X) \begin{bmatrix} b \\ m \end{bmatrix} = X^T . y$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} = \begin{bmatrix} 11 \\ 25 \end{bmatrix}$$

$3b + 6m = 11$ — (1)

$6b + 14m = 25$ — (2)

$2 \times 3b + 6m = 11$

$6b + 14m = 25$

$6b + 12m = 22$

$6b + 14m = 25$

$2m = +3$

$m = 3/2 = 1.5$

$m = 1.5,$ $\quad 3b + 6(m) = 11$

$3b + 6(1.5) = 11$

$3b + 9.0 = 11$

$3b = 11 - 9$

$3b = 2$

$b = 2/3 = 0.666$

(4)

$m = 1.5 \quad b = 0.666$

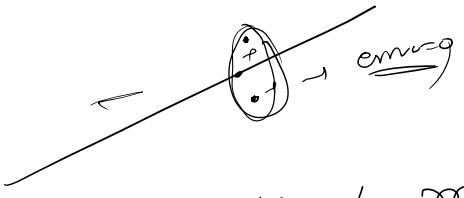
$y = 0.666 + (1.5)(x)$

$= 0.666 + (1.5)(4)$

$= 0.666 + 6.0 = \underline{\underline{6.666}}$

x	y	$\hat{y}$ prediction	error $y - \hat{y}$	SQerr
1	2	2.16	−0.16	0.0256
2	4	3.66	0.33	0.111
3	5	5.166	−0.163	0.0266

$MSE = 0.0256 + 0.111 + 0.0266/3 = \underline{\underline{0.053}}$



$\bar{x} = 1 + 2 + 3 = 6/3 = 2$

$\bar{y} = 2 + 4 + 5 = 11/3 = 3.667$

$y = b + mx$

$= m = \sum (x_i - \bar{x})(y_i - \bar{y}) \Big/ \sum (x_i - \bar{x})^2$

$= b = \bar{y} - m \cdot \bar{x}$

x	$x - \bar{x}$	y	$y - \bar{y}$
1	−1	2	−1.667
2	0	4	0.333
3	1	5	1.44

Stability

LinearRegression → target

continuous

1	2
2	4
3	5

matrices

Assignment:

KNeighborClassifier

X = np.array([[3,4], [2,3] [4,5] [6,3] [5,8]])

Y = np.array([1, 1, 2, 2, 3])

(x_1, y_1) (x_2, y_2)

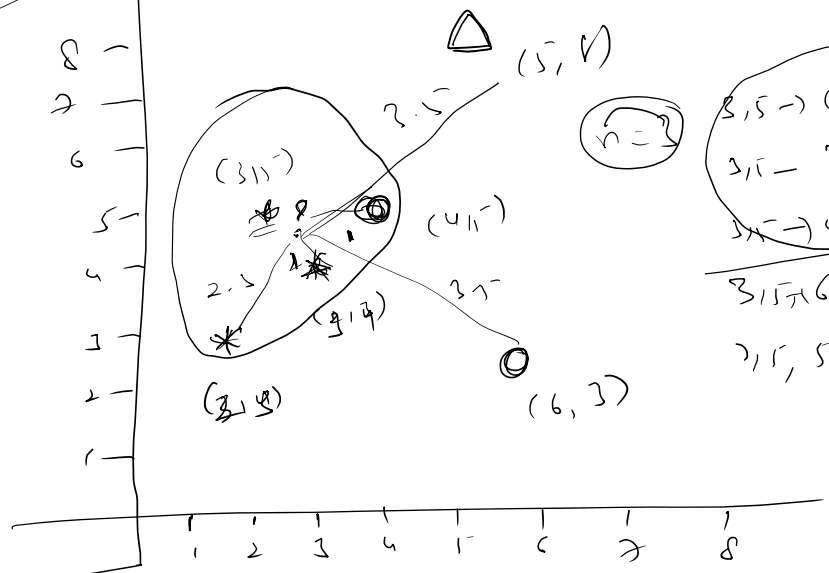
euclidean $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

P = [3,5]

KNN (n = 3)

odd number

3 — 2 ✗



n = 3

3,5 → (2,3) = 2.3

3,5 — 3,4 = 1

3,5 → 4,5 = 1

3,5 → 6,3 = 3.5

3,5, 5,8 = 3.5

$(3,5)(2,3) = \sqrt{(2-3)^2 + (3-5)^2} = \sqrt{1+2^2} = \sqrt{1+4} = \sqrt{5} = 2.3$

$(3,5)(3,4) = \sqrt{(3-3)^2 + (4-5)^2} = \sqrt{0+1} = \sqrt{1} = 1$

$(3,5)(6,3) = \sqrt{(6-3)^2 + (3-5)^2} = \sqrt{3^2+2^2} = \sqrt{9+4} = \sqrt{13} = 3.5$

$(3,5)(4,5) = \sqrt{(4-3)^2 + (5-5)^2} = \sqrt{1^2+0^2} = \sqrt{1} = 1$

$(3,5)(5,8) = \sqrt{(5-3)^2 + (8-5)^2} = \sqrt{2^2+3^2} = \sqrt{4+9} = \sqrt{13} = 3.5$

Euclidean $= \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

P(3,5)

$(3,4) = \sqrt{(3-3)^2 + (5-4)^2}$

$(2,3) = \sqrt{(2-3)^2 + (3-5)^2}$

$(4,5) = \sqrt{(4-3)^2 + (5-5)^2}$

$(6,3) = \sqrt{(6-3)^2 + (3-5)^2}$

$(5,8 = \sqrt{(5-3)^2 + (8-5)^2}$

2) Manhattan distance $= d = |x_2 - x_1| + |y_2 - y_1|$

$P(3,5)$

$(3,4) = |3-3| + |4-5| = 0 + 1$

$(2,3) = |2+3| + |3-5|$

$(4,5) = |4-3| + |5-5|$

$(6,3) = |6-3| + |3-5|$

$(5,8) = |5-3| + |8-5|$

Hamming Distance $= (3,4) \vee (3,5)$

$3 = 3 \quad 5 \neq 4 = 1$

$(2,3)$

2

Minkowski $\rightarrow d = \left(|x_2 - x_1|^p + |y_2 - y_1|^p \right)^{1/p}$

if $P = 1$ = $|x_2 - x_1| + |y_2 - y_1|$ ⟹ manhattan

if $P = 2$ = $\left(|x_2 - x_1|^2 + |y_2 - y_1|^2\right)^{1/2}$

$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ euclidean

, $P = 3$

algorithm → linear Regression → Regression

KNN — distance based → classification